FINANCIAL TRANSACTIONS & LOANS DASHBOARD MICROSOFT POWER BI PROJECT REPORT

NANDITA A

ROLL NO.: E25FA031

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TOOL USED: MICROSOFT POWER BI DESKTOP

1. INTRODUCTION

For this project, I created a Financial Transactions & Loans Dashboard using Microsoft Power BI Desktop. I worked with a banking dataset that contains 11 tables related to customers, accounts, loans, transactions, branches and other details.

My main aim was to organise the data and present it in a way that is easy to understand. I used KPI cards, charts, a table and slicers to show important information about customers, account activity, loans and transactions.

Through this project, I was able to practise the different steps involved in building a dashboard, from cleaning the data to creating visuals and adding interactive features.

2. OBJECTIVES

The main objectives of my project were:

- To study the customer, account, loan and transaction details in the dataset.
- To calculate important values such as Total Customers, Active Accounts, Active Loans and Total Transaction Value.
- To compare different account types and loan statuses.
- To understand how transaction activity changes over time.
- To use slicers so that users can filter and explore the dashboard.
- To apply drill-down, tooltips and conditional formatting to make the report more useful.

3. DATASET DESCRIPTION

The dataset I used contains the following 11 tables:

- Customers
- Customer Types
- Accounts
- Account Types
- Account Statuses
- Loans
- Loan Statuses
- Transactions
- Transaction Types
- Branches
- Addresses

These tables contain information such as customer details, account balances, loan amounts, interest rates and transaction records.

I connected the tables in the data model so that I could analyse related information together instead of working with each table separately.

4. DATA PREPARATION

Before creating the visuals, I cleaned and formatted the data using Power Query Editor.

The main steps I carried out were:

- Removed duplicate values from the Account ID, CustomerID, Address ID and LoanID columns.
- Changed the relevant date columns to the Date data type.
- Converted Balance, PrincipalAmount and Amount to the Fixed Decimal Number format.
- Replaced missing customer names and address details with “unknown”.
- Used the English (India) locale to correct the DOB format and replaced errors with null values.
- Applied Trim and Clean to the transaction descriptions to remove extra spaces and unwanted characters.

This helped me make the data more consistent before using it in the dashboard.

5. RELATIONSHIPS

After preparing the data, I created 12 relationships between the tables in Model view.

These relationships connected customer details with accounts, accounts with loans, and transactions with transaction types and branches.

I kept some relationships inactive, including the relationship between Accounts and Loans and the relationship between Accounts and Transactions through AccountDestinationID.

For calculations that needed these inactive relationships, I used USERELATIONSHIP() in the relevant DAX measures.

6. DAX MEASURES

I created DAX measures to calculate the main values needed for the dashboard.

- Total Customers: Counts the unique customers in the dataset.
- Total Accounts: Counts the distinct accounts.
- Total Account Balance: Adds the balances of all accounts.
- Active Accounts: Counts the accounts with an Active status.
- Total Loans: Counts the distinct loan records.
- Total Loan Principal: Adds the principal amounts of all loans.
- Average Interest Rate: Calculates the average interest rate.
- Active Loans: Counts the loans with an Active status.
- Total Transactions: Counts the unique transaction IDs.
- Total Transaction Value: Adds the transaction amounts.

I also created two additional measures, Destination Transaction Value and Total Loan Amount. I used USERELATIONSHIP() in these measures to carry out calculations through the inactive relationships.

7. VISUALS USED

7.1 KPI CARDS

I added four KPI cards at the top of the dashboard to show:

- Total Customers
- Active Accounts
- Active Loans

- Total Transaction Value

These cards give a quick overview of the main information without requiring users to go through all the charts.

7.2 ACCOUNTS BY ACCOUNT TYPE

I used a Clustered Column Chart to compare the number of accounts in each account category.

I placed Account Type on the X-axis and Total Accounts on the Y-axis.

This made it easier to see how account numbers differed across the categories.

7.3 TRANSACTIONS BY MONTH

I created a Line Chart using the Transaction Date hierarchy and Total Transactions.

This chart helped me examine how transaction counts changed over time and compare activity across different periods.

7.4 LOANS BY LOAN STATUS

I used a Donut Chart to show how loans were distributed across different statuses.

I selected Loan Status as the category and Total Loans as the value.

This provided a simple visual comparison of the number of loans under each status.

7.5 TRANSACTIONS BY TYPE

I created a Clustered Bar Chart using Transaction Type and Total Transactions.

This chart allowed me to compare the number of transactions recorded under each transaction category.

7.6 TRANSACTION SUMMARY

I added a Table visual with the following fields:

- Transaction Type
- Total Transactions
- Total Transaction Value

The table shows the transaction counts and values together, making it easier to compare the different transaction categories in detail.

8. SLICERS AND INTERACTIVITY

To make the dashboard interactive, I added three slicers:

- Year: To view transaction information for a selected year.
- Account Type: To filter the report by account category.
- Transaction Type: To view information for selected transaction categories.

I also used Format → Edit interactions to control how the visuals responded to selections.

Depending on the visual, I used the Filter, Highlight or None option to control how a selection affected the other visuals on the page.

9. DRILL-DOWN

I created a Line Chart called “Transactions Over Time” using the Transaction Date hierarchy on the X-axis and transaction counts on the Y-axis.

I used drill-down to explore the data at the year, quarter and month levels.

- Year Level: I viewed the yearly transaction activity and set the Y-axis range from 0 to 13,000.
- Quarter Level: I moved to the quarter level and adjusted the Y-axis range from 11,500 to 12,500 to examine quarterly activity.
- Month Level: I then moved to the month level and set the Y-axis range from 3,000 to 5,000 to study monthly changes.

I also used Drill Down mode to explore a particular period. I selected 2021, moved into Quarter 2 and then examined April, May and June.

This helped me move from a general view of transaction activity to a more detailed view of a selected period.

I captured screenshots of the year, quarter, month and selected data-point views to record the drill-down process.

10. REPORT-PAGE TOOLTIP

I created a separate page named “Transaction Tooltip” and enabled the tooltip option under Page information.

On this page, I added three cards to display:

- Transaction Type
- Total Transactions
- Total Transaction Value

I then linked the tooltip page to the Transactions by Type chart.

When a user hovers over a data point, the tooltip shows more information about that transaction category without taking them away from the main dashboard.

11. CONDITIONAL FORMATTING

I applied data bars to the Total Transaction Value column in the Transaction Summary table.

As required in the project instructions, I used green bars for positive values and red bars for negative values.

This made the values easier to compare visually and helped distinguish positive amounts from negative amounts.

12. KEY FINDINGS

The dashboard helped me examine:

- The total number of customers and accounts.
- The number of accounts and loans that have an Active status.
- The account category with the highest number of accounts.
- Changes in transaction counts over time.
- The number of loans under each loan status.
- Differences in transaction counts and values across transaction categories.

The exact numerical findings still need to be added after checking and validating the final results in Power BI.

CONCLUSION

By completing this project, I gained practical experience in using Microsoft Power BI to work with financial data. I practised cleaning data, creating relationships, writing DAX measures and designing an interactive dashboard.

The KPI cards provided a quick summary, while the charts and table helped me examine the data in more detail. I also used slicers, drill-down and tooltips to make the dashboard easier to explore.

Overall, this project helped me better understand data modelling and financial data analysis. It also gave me experience in turning raw data into a clear and organised report using Power BI.